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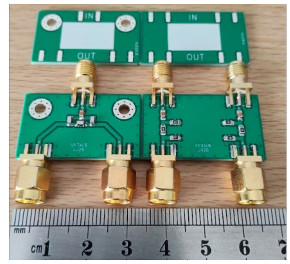
Version 1.2
30 August 2026



These adapter boards are intended to limit wear to the tinyGTC A and B connectors as well as reducing the possibility of damage to the GTC input circuits. They were intended for use at 10 MHz but are usable above 1 GHz. They may not be suitable for making precision level/amplitude measurements.

Please note: The use of these adapter boards does not eliminate the need to adhere to the tinyGTC manufacturer's instructions for input protection.

Splitter board

The splitter board provides 6dB attenuation and uses a 100nf DC block capacitor. The mounting holes are for use in other projects.

Attenuator board

The attenuator board uses 100nf DC block capacitors and has a Pi attenuator for each channel.

Design

The board size is 34mm x 20mm x 1.6mm and made from FR4 material. They use 0805 SMD components and low cost SMA connectors. The white painted area on the front side of the board can be written on with permanent maker.

The spacing of the A & B connectors on the tinyGTC is not specified but a survey of 3 x tinyGTC devices suggests it is likely 22mm +/- 0.5mm. HOWEVER, it is best to custom build each board using the tinyGTC as a template to ensure no stress is placed on the tinyGTC connectors.

Cost

Integrating a DC blocking capacitor makes these boards a cost effective solution compared to the purchase of individual SMA DC blocks and attenuators.

Disclaimer

No advice has been sought from the tinyGTC manufacturer in relation to the use of these boards.

These boards demonstrate my approach to using the tinyGTC and while my experience revealed no issues, you should consider there may be undiscovered errors.

Should you undertake a similar project, I will not accept responsibility for anything that goes wrong with your equipment.

Construction

WARNING!!

During the connector installation phase there is a risk that you can damage your tinyGTC either by applying too much mechanical force or too much heat.

There is also a small risk that a potential difference between the tip of the soldering iron and the shield of the SMA connectors could damage the input channels of the tinyGTC. Make measurements before you proceed.

Each board should be custom built to fit a specific tinyGTC.

Remain in control as you work. A sudden release of energy when fitting connectors could result in damage. Applying too much heat or spending too long soldering the connectors could result in damage.

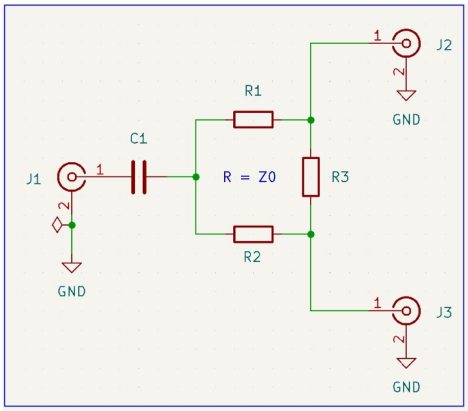
- Start with the splitter if you intend to use one because you can use it as an alignment jig when adding input connectors to the attenuator board.
- Begin by laying the tinyGTC face down on a soft cloth.
- Fit male SMA connectors to the A & B connectors on the tinyGTC.
- Align/rotate the SMA flanges so the PCB can be fitted to both connectors.
- Align the board so the centre pins of the connectors are symmetrical against the PCB pads and both the connector flange pins are resting on the PCB. The best pin location may not be in the centre of the solder pads.
- If necessary, use an alligator clip or some other mechanism to hold the connectors in place against the board.
- Working quickly, apply just enough solder to hold the centre pins to the board – do not solder the flange pins yet.
- Carefully remove the PCB assembly with connectors, check the alignment of the pins one last time then solder the flange pins and finally re-flow the centre pin.
- Complete installation of the remaining connectors in the same way.
- Add SMD components only after you're satisfied with the installation of <u>all</u> the connectors.
- Confirm the signal characteristics of the board(s) are as expected.

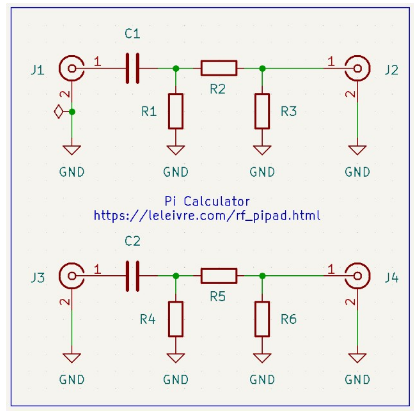
If all is well, you will have an adapter board that fits your tinyGTC without minimal mechanical stress.

Results

Input limitations of the tinyGTC should be considered when measuring unknown signals.

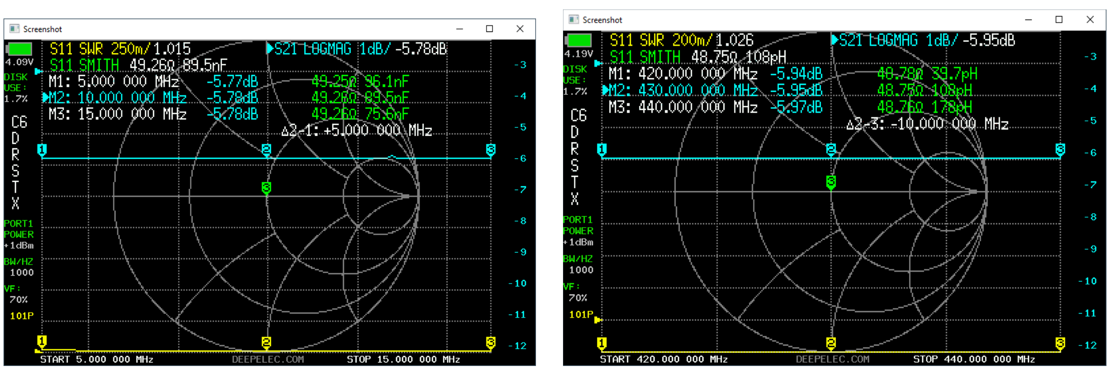
Please review the tinyGTC manufacturer's advice regarding input protection.



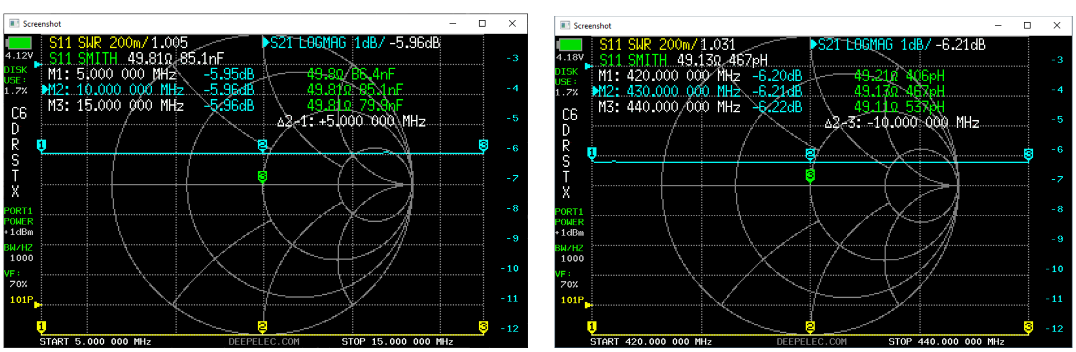


VNA test results at 10 MHz and 430 MHz

The boards were designed primarily to protect the tinyGTC, a broad frequency response was not considered mandatory. Nevertheless, it is interesting to see how the boards perform at higher frequencies.



Splitter board: 10 MHz and 430 MHz, second port terminated in 50 ohms.



Checking the 6dB attenuator board at 10 MHz and 430 MHz. Subsequent testing showed both boards delivered similar responses at 1296 MHz.

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